



## The 66th ASH Annual Meeting Abstracts

## POSTER ABSTRACTS

## 721.ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TOXICITIES

**Omitting Ciprofloxacin Prophylaxis in Allogeneic Hematopoietic Stem Cell Transplantation Does Not Result in Inferior Outcomes**

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**Introduction**

Infectious complications remain a leading cause of morbidity and mortality after allogeneic hematopoietic stem cell transplantation (alloSCT). Antimicrobial prophylaxis regimens often include fluoroquinolones (FQ), such as ciprofloxacin, to prevent bacteremia by decreasing oral and intestinal pathogens, which are common sites for bacterial translocation after mucosal barrier injury. However, the emergence of antibiotic stewardship programs and concerns about the promotion of antimicrobial resistance and *Clostridioides difficile* infections (CDI) have raised questions about the use of prophylactic FQ. Additionally, early disruption of microbiota and dysbiosis during alloSCT are linked to increased mortality and the incidence of Graft-versus-Host Disease (GvHD). Accordingly, our center discontinued antibiotic prophylaxis for patients undergoing alloHSCT in May 2019. The aim of this retrospective study was to analyze the effect of omitting FQ prophylaxis. We hypothesized non-inferiority regarding the incidence of bloodstream infections (BSI), GvHD rates, CDI and overall survival (OS).

**Methods**

All patients who underwent first alloHSCT between 2017 and 2022 at our center were included. A total of 292 patients were analyzed, 116 patients received FQ prophylaxis (FQ cohort) and 176 did not receive any antibiotic prophylaxis (non-FQ cohort). Patients were considered to have received antibiotic prophylaxis if ciprofloxacin was administered with 500 mg BID from the day of admission to the first day of fever. Chi-squared test was used to compare categorical data. Continuous variables were compared using the t-test. OS was estimated using the Kaplan-Meier method and compared using the Log-rank (Mantel-Cox) test.

**Results**

Baseline characteristics were well balanced in both groups; median age was 57 years (range 18-73) in the FQ group versus (vs) 54 years (18-74) in the non-FQ group ( $p=0.46$ ). Acute myeloid leukemia or myelodysplastic syndrome were the most common diseases with 55 (47%) vs 93 (53%) ( $p=0.51$ ) in the FQ and non-FQ groups, respectively. Fifty-one (44%) vs 92 (52%) patients were cytomegalovirus (CMV) positive. Male patients were more frequent in both groups (70 (60%) vs 104 (59%),  $p=0.83$ ). Most patients received a peripheral stem cell graft (102 (88%) vs 164 (94%),  $p=0.06$ ) and matched unrelated donors were most common (72 (62%) vs 107 (54%)  $p=0.17$ ). BSI rates were comparable in both groups (FQ: 48 (41%) vs non-FQ: 85 (48%), ( $p=0.25$ )) and the frequency of central-line associated BSI were also similar in both groups (17 (35%) vs. 22 (39%),  $p=0.25$ ). Notably, incidence of gram-positive or polymicrobial BSI was significantly higher in the FQ prophylaxis group (49 (85%) vs 56 (67%),  $p=0.01$ ). The incidence of infections from other sources of such as urinary tract infections (26 (22%) vs 38 (22%),  $p=0.87$ ) or pneumonia (38 (33%) vs 46 (26%),  $p=0.22$ ) showed no significant difference. CDI were rare but comparable in both cohorts (4 (3%) vs. 6 (3%),  $p=0.99$ ). The number of intensive care unit admissions during alloSCT was similar (15 (13%) vs 23 (16%),  $p=0.93$ ). Patients were hospitalized for a median of 36 days (range 16-113 vs 11-147) in both groups ( $p=0.55$ ). No difference in the rate of aGvHD of the gut was seen; grades I-II° were observed in 19 (16%) vs 38 (22%) ( $p=0.27$ ) and grades III-IV° in

24 (21%) vs 41 (23.3%) ( $p=0.60$ ) of FQ and non-FQ patients, respectively. Early transplant mortality at 60 days after alloSCT was low in both groups (6% vs 5%,  $p=0.60$ ). In a univariate analysis, one-year OS was significantly higher in the non-FQ group ( $p=0.02$ ). This was most likely due to the introduction of letermovir for CMV positive patients at our center in 2019, as the OS difference was only observed in the subgroup of CMV-positive patients ( $p=0.01$ ) and not in CMV-negative patients ( $p=0.44$ ).

#### Conclusions

Omitting FQ prophylaxis did not have a negative impact on OS, aGvHD of the gut or BSI and CDI incidence. The superior OS in the non-FQ cohort is most likely explained by confounding factors such as the introduction of letermovir in 2019. These data support our approach and may encourage other centers to omit FQ prophylaxis as part of antibiotic stewardship efforts.

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